

EXPERIENCE

GLOBAL ACTION ALLIANCE, Chicago, IL

Software Engineer (Data Engineer)

September 2024 - Present

- Created and maintained **ETL** pipelines using **Python**, for Clinical Trials WISE-R and Health WISE-R applications, by streamlining data ingestion and transformations to load into MongoDB. These pipelines increased productivity by 80%.
- Designed a document tagging algorithm leveraging **BioBERT** to map to ICD-11 WHO schema, improving clinical trial disease classification accuracy to **95%**.

GLOBAL ACTION ALLIANCE, Chicago, IL

Software Engineer (Data Engineer)

July 2023 – August 2023

- Developed and implemented **Python**-based web scraping algorithms using **Selenium**, capturing over 90% of available clinical trial records from 80+ countries, saving **3+ hours** of manual work daily.
- Collaborated with cross-functional teams to build and optimize machine learning models, leveraging processed data to generate insights and enhance decision-making processes.

CARELON (formerly Legato Health Technologies), Bengaluru, Karnataka, India

Associate Software Engineer

June 2021 – July 2022

- Automated the data verification pipeline for health insurance data using **Java**, **SQL** and **MongoDB**, reducing data entry errors by **65%** and increasing **processing speed** by **90%**.
- Strengthened system security during **AWS migration** by updating dependencies, mitigating **85% of vulnerabilities**, ensuring robust codebase stability.
- Designed and deployed backend services using **Java** and **Spring Boot**, optimizing system performance and scalability.

PROJECTS

RESEARCH PROJECT ON BLOCKCHAIN

August 2023 – May 2024

- Created a **power-efficient Proof of Space algorithm** using **C++**, **BLAKE3 cryptographic hash function** and persistent storage, achieving up to **7x faster hash generation** and **99% lower energy consumption**.
- Utilized **CUDA** and **Pthreads** for multi-threaded sorting and binary search, enhancing performance up to **3.3x over Chia** across various hardware configurations.

GAS PUMP SYSTEM DEVELOPMENT

January 2023 – May 2023

- Designed and implemented a scalable gas pump system in **Java**, incorporating **multi-threading** for real-time operations and payment **API integration**.
- Conducted comprehensive testing, achieving **89% code coverage** and a **cyclomatic** complexity as low as 8 making this a high-quality industry level software.

REAL TIME MAP UPDATING BY SELF DRIVING CAR ([IJESC Paper](#))

July 2019 – December 2019

- Formulated a pioneering research study on integrating **self-driving vehicles** with **cloud computing**, utilizing sensor data for real-time map updates.
- Developed a paper outlining the benefits and challenges of **real-time map updates** in self-driving cars using cloud computing, highlighting the potential for **improved traffic predictions** and enhanced **emergency response times**.

EDUCATION

ILLINOIS INSTITUTE OF TECHNOLOGY, Chicago, IL

Degree: **Master of Science, Computer Science**

May 2024

Graduate Teaching Assistant, Advanced operating System

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, Chennai, Tamil Nadu, India

Degree: **Bachelor of Technology, Computer Science and Engineering**

July 2021

SKILLS

- Programming Languages:** Python, C, C++, Java, SQL, React, Typescript.
- Framework and Libraries:** Spring Boot, Selenium, Pandas, NumPy, Scikit-learn, TensorFlow, Maven.
- Tools:** MongoDB, Docker, Jenkins, Kubernetes, Tableau, Hadoop, Spark, Git, Shareflow, Databricks.
- Cloud Platforms and Domains:** AWS, Google Cloud Platform, ETL Pipelines, Distributed Programming, Blockchain, ML/AI, Parallel Programming.
- Google Certification:** Architecting with Google Compute Engine